$\boldsymbol{\theta}$

Forward pass

IPM Solver

$K_i \Delta \bar{z}_i = b_i$

Generalized Riccati Factorization

Retained Riccati Cache C_F

Nominal solution

$\boldsymbol{z^*(\theta)}$

Target adjoint system

$K_B^T a^* = g$

$\boldsymbol{g}$

Initialization

Cached candidate

$a_0 = S_F(g)$

Residual check

$\rho_0 \leq \varepsilon$

Yes

No

Preconditioning

GMRES correction if needed

Backward pass

$\boldsymbol{d\mathcal{L}/d\theta}$